

Advisor:

Dr. Karen Mulleners

Supervisor:

Dr. Karen Mulleners

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Influence of pneumatic tube length on pressure measurements

Thomas Lepère

EPFL • STI • SGM

Abstract Wall-pressure measurements with remote sensors are affected by the pneumatic line that connects the pressure tap to the transducer. This effect is critical for unsteady measurements, because the tube can attenuate and phase-shift pressure fluctuations before they reach the sensor. This study quantifies the influence of tube length on pressure measurements around a circular cylinder at $Re \simeq 4.0 \times 10^4$. A single wall pressure tap is connected to a differential pressure sensor through tubes with lengths between $L = 300 \text{ mm}$ and $L = 2000 \text{ mm}$, and the cylinder is rotated to sample angular positions between $\alpha = 0^\circ$ and $\alpha = 180^\circ$. The mean pressure coefficient reproduces the expected separated-flow distribution and remains weakly affected by tube length. The fluctuating pressure, in contrast, decreases systematically as the tube length increases. The standard deviation is reduced by approximately 36 % for the longest tube relative to the shortest tube. Spectral analysis shows that this attenuation is frequency dependent and that the energetic spectral band changes with tube length instead of collapsing onto a single nominal shedding frequency. This behaviour suggests that the measured spectra result from the interaction between aerodynamic pressure fluctuations and the acoustic response of the pneumatic tube. Tube length has a limited effect on mean pressure measurements but must be accounted for when unsteady pressure amplitudes or spectral peaks are interpreted.

Introduction

Wall pressure taps are commonly used in wind tunnels because they measure surface pressure with minimal disturbance to the external flow (Bynum, Ledford, and Smotherman, 1970). A pressure tap is a small orifice drilled into the model surface and connected to a pressure sensor by a pneumatic tube. The sensor output depends on two quantities: the aerodynamic pressure at the tap and the dynamic response of the tube-sensor system (Taback, 1949).

The dynamic response becomes important when the pressure field is unsteady. The air contained in the tube behaves as a pneumatic transmission line whose response depends on the tube length, tube diameter, tube material, fluid properties, and sensor volume. Pressure fluctuations may then be attenuated, amplified near resonant frequencies, and phase shifted before they are recorded by the sensor. Bajsic, Kutin, and Zagar (2007) modeled this response as a single-degree-of-freedom fluid oscillator similar

to a Helmholtz resonator. Their step-response measurements showed that increasing the tube length decreases the resonant frequency and increases the response time and damping. Sinclair and Robins (1952) similarly showed that the response time of capillary pressure-measurement systems increases with tube length and gas viscosity, and decreases strongly with tube internal diameter.

Frequency-response measurements provide the same physical picture in the spectral domain. Wang et al. (2018) measured the ratio between inlet and outlet pressure spectra for polyvinyl-chloride tubes with an inner diameter of 1.2 mm and lengths up to 1400 mm. The tube system first amplifies and then attenuates the pressure fluctuations as frequency increases, while the phase lag increases with tube length. The modal peak frequencies decrease with tube length, which is consistent with a one-dimensional acoustic interpretation of the gas column.

This limitation is relevant for the experimental validation of aerodynamic simulations performed by the EPFL Racing Team. Wind-tunnel tests on a scaled race car model compare measured aerodynamic loads and wall pressures with numerical predictions. The pressure sensors cannot always be placed directly behind the pressure taps, so the tubes that connect the taps to the sensors may bias unsteady measurements. Quantifying this bias helps determine whether differences between experiments and simulations come from the flow physics or from the measurement chain.

The objective of this study is to quantify the influence of pneumatic tube length on wall-pressure measurements. A circular cylinder is used as a reference geometry because its mean pressure distribution and vortex-shedding frequency are well documented (Schiller and Linke, 1933; Lienhard, 1966). A single pressure tap is connected to a differential pressure sensor through tubes of different lengths, and measurements are repeated at several angular positions. The analysis focuses on the mean pressure coefficient, the fluctuation level, and the spectral content of the pressure signal.

Methods

Experimental setup

The test bench consists of a vertical circular cylinder of diameter $D = 38$ mm and height $h = 248$ mm. The cylinder is mounted on a star-shaped support (fig. 1b) that sets the angular position of the pressure tap in increments of 30° . The pressure tap is located $l = 41$ mm below the top of the cylinder. The top of the cylinder is positioned $h_o = 18$ mm below the wind-tunnel outlet, and the cylinder axis is located $d = 12$ mm downstream of the outlet plane. The outlet has a square cross-section of side length 140 mm.

The pressure tap is connected to a *Honeywell SSCMRRD010MD2A3* differential pressure sensor through a pneumatic tube of length L (fig. 1). The sensor measures the pressure difference $p - p_\infty$, where p is the pressure transmitted from the tap and p_∞ is the outlet static pressure. Tests are carried out at a freestream velocity $U_\infty = 14.2 \text{ m s}^{-1}$. The air temperature is $T = 25.4^\circ\text{C}$ and the atmospheric pressure is $p_{\text{atm}} = 1016.6 \text{ hPa}$. The Reynolds

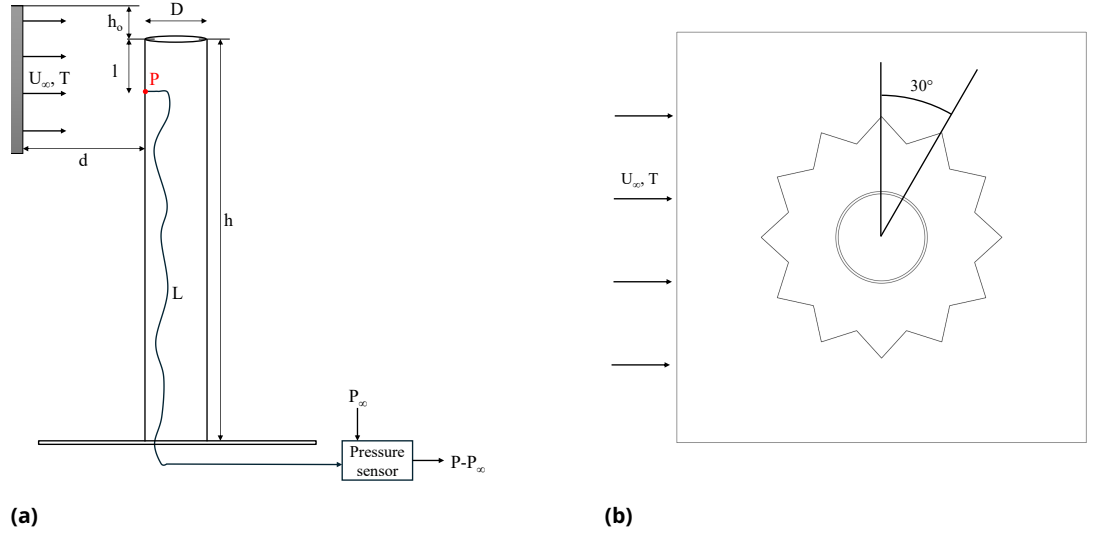


Figure 1. Test bench used to measure the response of the pneumatic tube system. (a) Side view of the cylinder subjected to a freestream velocity U_∞ and connected to the differential pressure sensor through a tube of length L . (b) Top view of the indexing system used to vary the angular position α of the pressure tap relative to the incoming flow.

number based on the cylinder diameter D is

$$Re = \frac{\rho U_\infty D}{\mu} \simeq 4.0 \times 10^4, \quad (1)$$

where U_∞ is the upstream velocity, ρ is the air density and μ is the dynamic viscosity of air.

At this Reynolds number, the expected Strouhal number of a circular cylinder is approximately $St = 0.2$ (Lienhard, 1966). This value corresponds to a vortex-shedding frequency of approximately $f = St U_\infty / D = 75 \text{ Hz}$ for the measured freestream velocity. The pressure signal is sampled at

$$f_s = 1000 \text{ Hz}, \quad (2)$$

which satisfies the Shannon–Nyquist criterion for the expected shedding frequency (Shannon, 1949). Each acquisition lasts 40 s to ensure the statistics (mean, eq. (8) and standard deviation, eq. (9)) convergence.

The angular positions tested are

$$\alpha \in \{0, 30, 60, 90, 120, 150, 180\}^\circ, \quad (3)$$

and the tube lengths are

$$L \in \{300, 400, 500, 600, 700, 800, 900, 1000, 1200, 1400, 1600, 1800, 2000\} \text{ mm}. \quad (4)$$

Non-dimensional quantities

The cylinder diameter D is used as the reference length and the freestream velocity U_∞ as the reference velocity. The pressure coefficient is defined as

$$C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U_\infty^2}, \quad (5)$$

where $p - p_\infty$ is the differential pressure measured by the sensor, and ρ the air density. The inviscid pressure distribution around a cylinder in potential flow is used as a reference (Batchelor, 2000):

$$C_{p,\text{pot}}(\theta) = 1 - 4 \sin^2 \theta, \quad (6)$$

where θ is the angular position measured from the stagnation point. The Strouhal number associated with a frequency f is

$$St = \frac{f D}{U_\infty}. \quad (7)$$

Statistical analysis

The wake of the cylinder is unsteady at the Reynolds number considered, so the pressure signals are compared through statistical quantities rather than through pointwise time traces. For a discrete signal x_i containing N samples, the time average is

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad (8)$$

and the fluctuation level is quantified by the standard deviation

$$\sigma_x = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}. \quad (9)$$

Because pneumatic tubes can damp pressure fluctuations (Bajsic, Kutin, and Zagar, 2007), a decrease in σ_{C_p} with increasing L is interpreted as attenuation of the unsteady pressure component.

Spectral analysis

The frequency content of the pressure fluctuations is analyzed using the discrete Fourier transform (Cooley and Tukey, 1965; Press et al., 2007). For a signal sampled at frequency f_s , the frequency associated with Fourier mode k is

$$f_k = \frac{k}{N \Delta t} = \frac{k f_s}{N}, \quad (10)$$

where $\Delta t = 1/f_s$ is the sampling interval. The power spectral density is estimated using Welch's method (Welch, 1967). The signal is divided into M segments of length N_s . Each segment is multiplied by a window w , and the spectra are averaged:

$$\text{PSD}(f_k) = \frac{1}{M} \sum_{m=1}^M \frac{2}{f_s N_s} \left| \text{FFT}(x_m w)_k \right|^2. \quad (11)$$

Welch averaging reduces the variance of the spectral estimate and allows the energy distribution of pressure fluctuations to be compared across tube lengths.

The relative spectral gain quantifies the frequency-dependent effect of tube length:

$$G(f, L) = 10 \log_{10} \left(\frac{\text{PSD}(f, L)}{\text{PSD}(f, L_0)} \right), \quad (12)$$

where $L_0 = 300$ mm is the shortest tube length. Values $G < 0$ indicate attenuation relative to the reference tube.

Results and Discussion

Mean pressure distribution

The mean pressure distribution around the cylinder first confirms that the experiment captures the expected aerodynamic behaviour. The mean pressure coefficient is close to unity at $\alpha = 0^\circ$, which corresponds to the stagnation region (fig. 2a). It decreases rapidly with angular position and reaches its most negative values at $\alpha = 60^\circ$. Downstream of this region, $\overline{C_p}$ remains approximately constant around -1 , as expected in the separated wake of a circular cylinder at $Re \simeq 4.0 \times 10^4$ (Schiller and Linke, 1933; Lienhard, 1966).

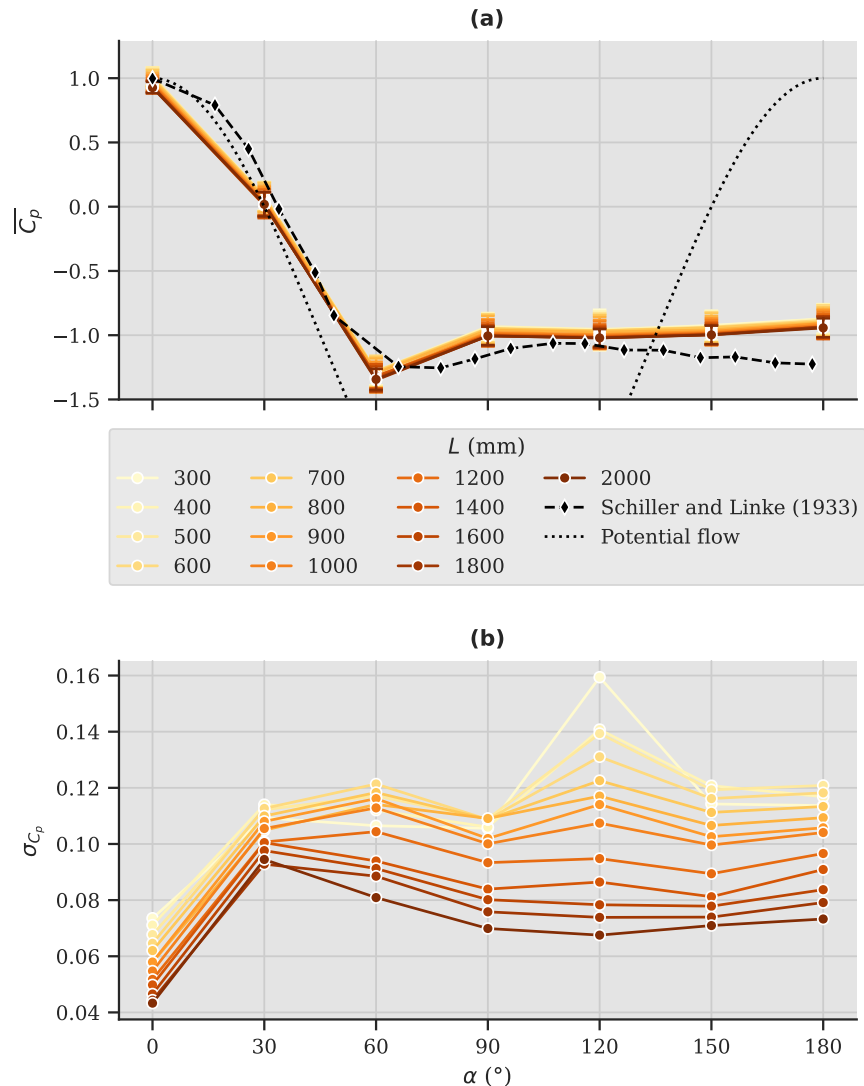


Figure 2. Mean pressure coefficient and pressure-fluctuation level as functions of the angular position of the pressure tap. (a) The measured mean pressure coefficient is compared with the experimental reference of Schiller and Linke (1933) and with the inviscid potential-flow prediction. (b) The standard deviation of the pressure coefficient decreases as the tube length increases.

The measured mean distribution follows the experimental reference of Schiller and Linke (1933) more closely than the inviscid potential-flow prediction. Potential flow captures

the stagnation pressure at $\alpha = 0^\circ$, but it predicts pressure recovery on the rear side of the cylinder. This recovery is absent because the real flow separates from the cylinder surface. The agreement with the experimental reference indicates that the tap and sensor system capture the mean flow topology despite the finite tube length.

Tube length has only a weak influence on $\overline{C_p}$. At each angular position, the curves obtained with different values of L remain close to each other compared with the angular variation of the mean pressure. The main pressure distribution is not strongly biased by the pneumatic tube length over the range investigated here. Small deviations between tube lengths remain visible in the separated-flow region, where the signal contains stronger unsteady components.

Pressure fluctuations

The fluctuation level is more sensitive to tube length than the mean pressure. The standard deviation σ_{C_p} decreases systematically as L increases (fig. 2b). This trend appears at all angular positions and is strongest where the unsteady pressure fluctuations are largest. The smallest fluctuations occur near the stagnation region, while larger fluctuations occur for $\alpha \geq 30^\circ$ because shear-layer motion and vortex shedding dominate the pressure signal. The systematic decrease of σ_{C_p} shows that the tube acts mainly as a dynamic damping element: it leaves the mean pressure nearly unchanged but attenuates the measured unsteady component.

At $\alpha = 120^\circ$, the pressure fluctuations reach their largest level for the shortest tubes, which indicates that this angular position is particularly sensitive to the unsteady wake dynamics. This region is located on the rear side of the cylinder, where vortex shedding and wake motion produce strong pressure variations. As the tube length L increases, σ_{C_p} decreases and the difference between successive tube lengths becomes progressively smaller. This behaviour suggests that the tube first removes a large fraction of the high-frequency fluctuation energy, while further increases in L mainly affect an already attenuated signal. The attenuation tends to saturate for the longest tubes: the pneumatic line still damps the unsteady pressure component, but its marginal effect on σ_{C_p} becomes weaker.

The relative bias in σ_{C_p} compares the fluctuation level measured with each tube to the reference length $L_0 = 300$ mm. A positive value means that the tube transmits stronger fluctuations than the reference tube, while a negative value means that the tube attenuates the fluctuations. The bias remains close to zero for the shortest tubes and then decreases almost monotonically as L increases (fig. 3). The attenuation reaches approximately 36 % for $L = 2000$ mm.

This quasi-monotonic attenuation is consistent with the expected response of a pressure-measurement system that contains a long pneumatic line (Bajsic, Kutin, and Zagar, 2007; Sinclair and Robins, 1952). The tube length increases the inertia and viscous losses of the air column, which reduces the amplitude of pressure oscillations transmitted to the sensor. The bias is small for mean quantities but important for unsteady quantities.

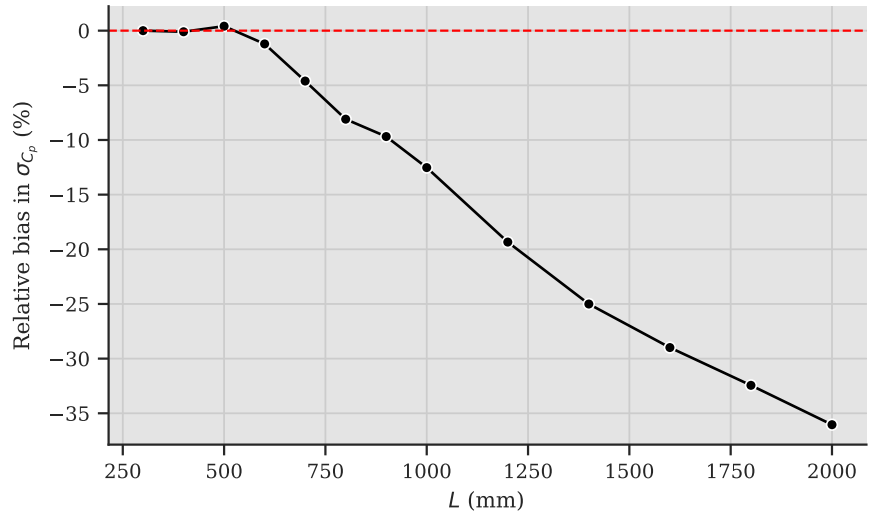


Figure 3. Relative bias in the pressure-coefficient standard deviation with respect to the shortest tube length $L_0 = 300$ mm. The increasing negative bias indicates that longer tubes attenuate the measured pressure fluctuations.

Spectral analysis

The spectra measured at $\alpha = 150^\circ$ reveal a frequency-dependent attenuation of the pressure fluctuations (fig. 4a). In the low-frequency range, here defined as $St \lesssim 0.1$, the spectra obtained with different tube lengths remain close to each other. In the intermediate range, $0.1 \lesssim St \lesssim 0.5$, the spectral energy depends strongly on L . The shortest tubes show an energetic band near the expected shedding range, whereas longer tubes modify both the amplitude and the position of this band. For $St \gtrsim 0.5$, the spectral level decreases for all tube lengths and the spectra become progressively dominated by the attenuation imposed by the pneumatic line.

The pressure spectrum at $\alpha = 150^\circ$ does not exhibit a fixed dominant peak at the expected vortex-shedding frequency for all tube lengths. Vortex shedding is expected near $St \simeq 0.2$, which corresponds to approximately 75 Hz for the measured freestream velocity. The absence of a systematic peak at this frequency indicates that the recorded spectrum is not only governed by the aerodynamic excitation. Instead, the measured signal results from the coupling between the flow-induced pressure fluctuations and the frequency response of the pneumatic tube-sensor system.

The shift of the energetic peak with L is consistent with the modal response of the air column inside the tube. The one-dimensional wave equation predicts standing acoustic modes whose natural frequencies scale approximately as $f_n \propto nc/L$, where c is the speed of sound, L is the effective tube length, and n is the mode number (D'Alembert, 1750; Pierce, 2019). Increasing the tube length lowers the modal frequencies and changes the frequency bands in which the tube amplifies or attenuates pressure fluctuations. The modal peak frequencies observed in pneumatic tubes are also reported to decrease with tube length (Wang et al., 2018).

The dominant peak in fig. 4 should be interpreted as a coupled aerodynamic-acoustic

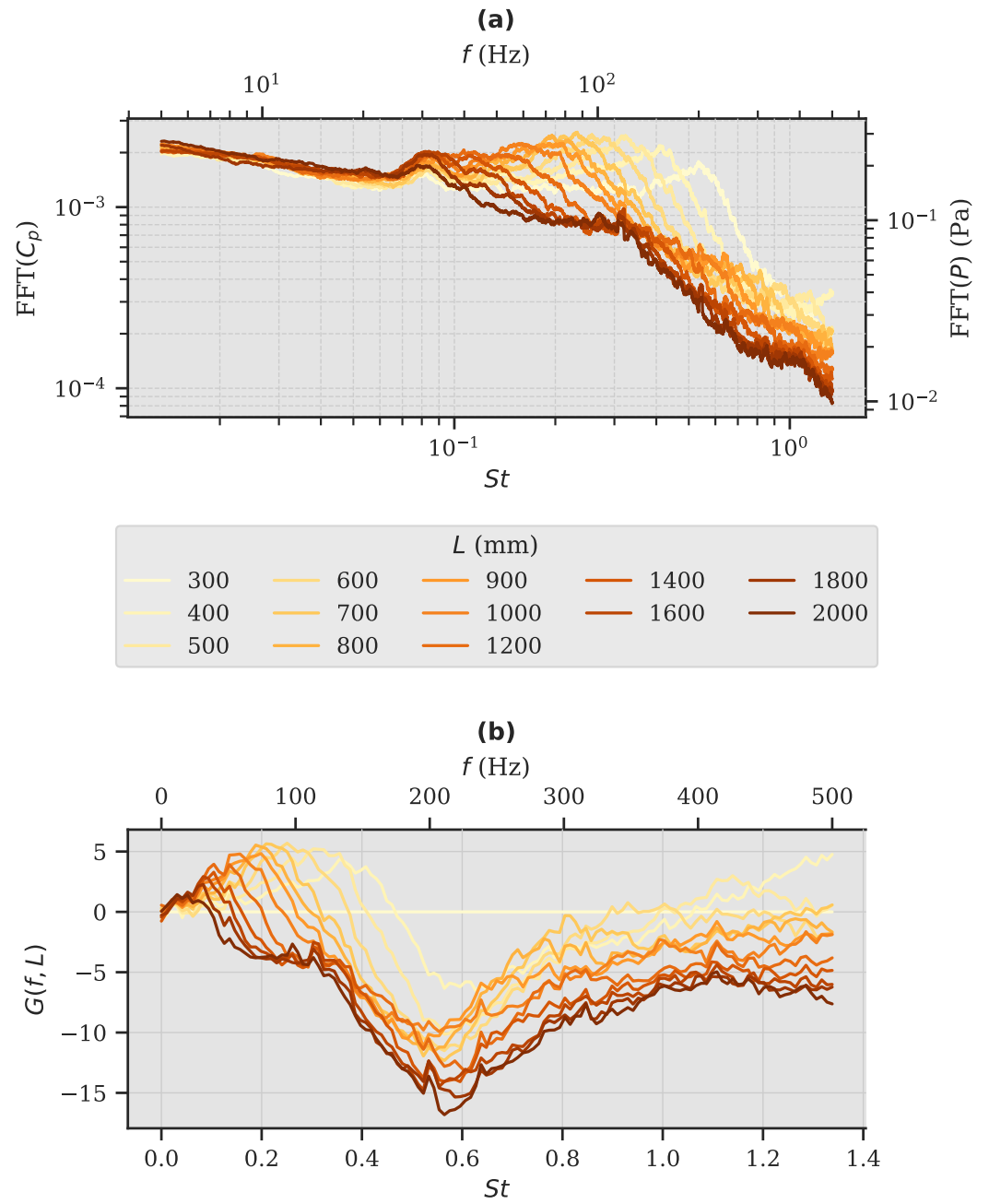


Figure 4. Spectral content of the pressure signal measured at $\alpha = 150^\circ$ and relative spectral gain with respect to the shortest tube. (a) The pressure spectra show a broad energetic band whose position and amplitude depend on tube length. (b) The relative spectral gain shows alternating amplification and attenuation, indicating that the tube response is frequency dependent rather than purely dissipative.

feature rather than as a direct measurement of the cylinder shedding frequency. The aerodynamic shedding supplies energy near $St \simeq 0.2$, but the pneumatic line modifies this energy through length-dependent resonances, anti-resonances, viscous losses, and end conditions. When a tube mode lies close to the shedding band, the sensor can record a maximum that is shifted away from the true shedding frequency. This explains why the peak position changes with L , even though the external flow conditions remain unchanged.

The relative spectral gain makes this coupling more explicit (fig. 4b). Some tube lengths produce a positive gain at low to intermediate Strouhal numbers, while all tube lengths show a pronounced negative gain around $St \simeq 0.5$ – 0.6 . This valley indicates strong attenuation over a band that shifts and deepens with tube length. At higher Strouhal numbers, the gain partially recovers but remains generally negative for the longest tubes. The tube cannot be represented by a constant attenuation factor. Its effect depends on frequency and may even include local amplification near resonant bands.

The angularly averaged gain confirms the overall attenuation observed in the standard deviation (fig. 5). The mean gain decreases as L increases for all angular positions, with the longest tubes reaching values between approximately -7 dB and -10 dB. The error bars indicate that the frequency-dependent response varies with angular position, but the trend with tube length remains consistent. Longer tubes reduce the spectral energy transmitted to the sensor, especially in the frequency range that contributes most to the pressure fluctuations.

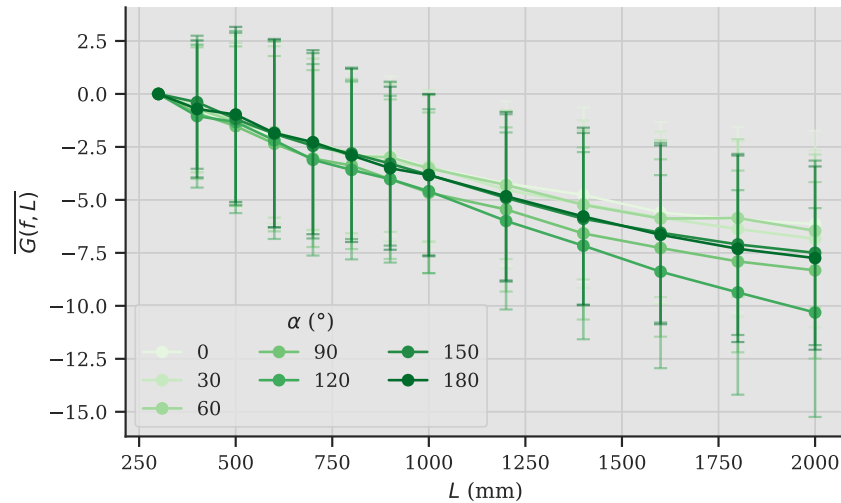


Figure 5. Angularly averaged spectral gain as a function of tube length. The monotonic decrease of $G(f, L)$ indicates that longer pneumatic tubes attenuate the unsteady pressure signal on average, even though local amplification can occur in specific frequency bands.

These spectral results explain why the standard deviation decreases with tube length. The tube removes part of the spectral energy carried by the unsteady pressure signal, and this loss accumulates over the frequency range used to compute σ_{C_p} . Mean pressure measurements are consequently comparatively robust to tube length, whereas spectral peaks and fluctuation amplitudes require either a transfer-function correction or a con-

sistent tube length across all measurements.

Conclusion

This study quantifies the effect of pneumatic tube length on wall-pressure measurements around a circular cylinder at $Re \simeq 4.0 \times 10^4$. The measured mean pressure coefficient reproduces the expected separated-flow distribution and agrees with the experimental reference more closely than with the inviscid potential-flow prediction. The mean pressure is only weakly affected by tube length over the range $L = 300$ mm to $L = 2000$ mm.

The unsteady pressure signal is much more sensitive to the tube length. The pressure-coefficient standard deviation decreases almost monotonically with L , reaching an attenuation of approximately 36 % for the longest tube relative to the shortest one. Spectral analysis shows that this attenuation is not uniform in frequency. The spectra contain broad energetic bands whose position and amplitude depend on L , and the dominant peak does not remain fixed at the nominal vortex-shedding frequency. This behaviour is consistent with a coupled aerodynamic-acoustic response: vortex shedding excites the pressure field, while the pneumatic tube modifies the transmitted signal through its length-dependent resonances and losses.

The tube length has a limited impact on time-averaged pressure coefficients but a significant impact on pressure fluctuations and spectral interpretation. Future measurements should either keep the pneumatic length identical for all taps or determine the transfer function of each tube-sensor system to correct the measured spectra and recover the pressure fluctuations at the tap.

References

- Bajsic, Ivan, Joze Kutin, and Tomaz Zagar (2007). "Response Time of a Pressure Measurement System with a Connecting Tube". In: *Instrumentation Science & Technology* 35.4, pp. 399–409.
- Batchelor, G. K. (2000). *An Introduction to Fluid Dynamics*. 1st ed. Cambridge University Press.
- Bynum, D. S., R. L. Ledford, and W. E. Smotherman (1970). *Wind Tunnel Pressure Measuring Techniques*: Fort Belvoir, VA: Defense Technical Information Center.
- Cooley, James W. and John W. Tukey (1965). "An algorithm for the machine calculation of complex Fourier series". In: *Mathematics of Computation* 19.90, pp. 297–301.
- D'Alembert, Jean Le Rond (1750). "Recherches sur la courbe que forme une corde tendue mise en vibrations". In: *Histoire de l'Académie royale des sciences et belles-lettres de Berlin pour l'année 1747*, pp. 214–219.
- Lienhard, John H. (1966). *Synopsis of lift, drag, and vortex frequency data for rigid circular cylinder*. Washington State University.
- Pierce, Allan D. (2019). *Acoustics: An Introduction to Its Physical Principles and Applications*. Cham: Springer International Publishing.
- Press et al. (2007). "Numerical Recipes: The Art of Scientific Computing". In: *Numerical Recipes: The Art of Scientific Computing*.

- Schiller, L. and W. Linke (1933). *Pressure and Frictional Resistance of a Cylinder at Reynolds Numbers 5,000 to 40,000*. Report 715. Number: NACA-TM-715. NACA.
- Shannon, C.E. (1949). "Communication in the Presence of Noise". In: *Proceedings of the IRE* 37.1, pp. 10–21.
- Sinclair, Archibald R and A Warner Robins (1952). "G A METHOD FOR THE DETERMINATION OF THE TIME LAG IN PRESSURE MEASURING SYSTEMS INCORPORATING CAPILLARIES". In.
- Taback, Israel (1949). *The Response of Pressure Measuring Systems to Oscillating Pressures*. Virginia, USA: NACA.
- Wang, Xue et al. (2018). "Effects of tube system and data correction for fluctuating pressure test in wind tunnel". In: *Chinese Journal of Aeronautics* 31.4, pp. 710–718.
- Welch, P. (1967). "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms". In: *IEEE Transactions on Audio and Electroacoustics* 15.2, pp. 70–73.

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